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Which Cause Is It? Selectively Blind Residuals for Isolating Compute Faults, Sensor Glitches, and Novel Classes in Multi-Detector Neural Inference

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Catalog counts are from the High Energy Astrophysics Science Archive Research Center (HEASARC) `fermigtrig` table on 17 September 2026. The five-seed tables use a simulator. Public continuous-time cubes and time-tagged event files are scored with the same lookup.

ABSTRACT A classifier on a redundant detector array can fail in four ways: a known source, an unseen class, a detector glitch, or silent corruption of its arithmetic. Open-set scores and range checks read activation magnitude, so a new physical class and a flipped weight can raise the same alarm. This paper specifies a four-symbol isolability dictionary, known, novel, sensor, and compute, and four residuals that are each blind to at least one symbol. A physics residual compares detectors and does not read the weights. A checksum of the linear head does not read the sky. Morphology on the shared profile cannot be invented by a corrupt head. The lookup reads the checksum first. Five Fermi Gamma-ray Burst Monitor simulator seeds give 90.0% four-way accuracy against 43.1% for a range check. Held-out flashes are novel in 95.0% of trials and a compute fault in 5.0%. A rounding bound removes that leak at 32-bit precision. A labeled fusion of the same residuals reaches 96.5% and usually calls a spike plus a bit-flip a sensor fault, where the lookup calls it compute. On 100 public continuous-time triggers, bit-flips stay isolated and no flash is called novel. A second draw keeps both results. Sensor isolation moves from 65% to 7.5% with direction. A Poisson likelihood ratio on one detector, amplitude maximized and with no detector response, raises those two rates to 100% and 55%. The same bit calls 20.6% of simulator flashes a sensor fault. At one millisecond, 23 of 24 flashes fall outside the burst morphology, and the lookup calls them sensor.

INDEX TERMS Silent data corruption, open-set recognition, Fermi GBM, structured residuals, isolability, algorithm-based fault tolerance, embedded neural networks

I. INTRODUCTION

A classifier on a detector array can need four different responses to one unusual activation: accept a known event, preserve a possible discovery, isolate a bad sensor, or recompute a flipped weight. Treating those outcomes as one anomaly is the error this paper is about. Adapting a model to a bit-flip poisons a good network. Discarding a new class as a glitch throws away the event.

The Fermi Gamma-ray Burst Monitor (GBM) makes the four causes concrete. Twelve sodium iodide (NaI) detectors and two bismuth germanate (BGO) detectors observe the unocculted sky. Catalogued triggers include gamma-ray bursts (GRBs), solar flares, magnetar bursts (SGRs), terrestrial gamma-ray flashes (TGFs), and local and distant particle events [1]–[3], [5], [6]. Public classifiers sort known classes and flag outliers [13]. Ground pipelines assume correct arithmetic. Onboard inference cannot. Silent data corruption is documented in production fleets [39]–[41], and soft errors can reach neural-network outputs [36], [37].

Open-set scores and range checks still collapse a new class and a bit-flip into one alarm [27], [30]–[33]. Classical isolability has no symbol for a new class [42]–[44]. Fermi pipelines veto single-detector spikes without an execution checksum [7], [9], [12].

Attribution by Residual-Invariance Analysis (ARIA) is a four-symbol dictionary, known, novel, sensor, and compute, read from residuals that are each blind to at least one symbol. The five-seed tables use a cosine-response simulator. Section V repeats the lookup on 100 public continuous-time (CTIME) cubes. Section VI-F scores time-tagged event

(TTE) light curves. Section VI reports the splits, the empty novel symbol, the spike scores, and the compound faults. Software bit-flips are not a beam test.

II. RELATED WORK

Table 1 places the closest methods against the four causes. A "part" entry means the method touches that column only inside an undifferentiated score. The isolability claim holds under the single-fault assumption only.

Open-set scores reduce a surprising input to a reject bit: maximum softmax probability [15], the energy score [16], ODIN [17], the Mahalanobis score [18], generalized ODIN [19], and virtual-logit matching [20]. OpenMax [21], the line that begins with Scheirer and colleagues [22], [23], and a strong closed-set classifier [24] add an unknown class. Gavarini and colleagues then treat those bits as fault detectors [27]. Selective classification still accepts or rejects and does not name a compute fault [28], [29]. Anomaly surveys and OpenOOD compare detectors inside that one class [25], [26]. ProGIP separates in-distribution inputs, out-of-distribution inputs, and high-order bit-flips with range checks on gradients [30]. ReAct showed that out-of-distribution inputs already occupy that extreme range [31].

Dr. DNA, quantile monitors, FACER, and BLINK flag silent corruption from activations or batch-normalization statistics, without a new-class symbol [32]–[35]. Fault injection measures how often a flip changes an output [36]–[38]. Structured parity residuals are insensitive to one fault and sensitive to another [42]–[45]. Algorithm-based fault tolerance detects a matrix-multiply error independently of the input [46], [47]. Intensity-guided and approximate checksums decide where to check and do not name discovery [48], [49]. The rounding bound below uses the standard model of floating-point error [50], [51]. On GBM, flight software requires a rate increase in two or more NaI detectors [1], [7], ground localization compares detector rates [8], [10], and a Poisson matched filter searches every detector [12]. Convolutional-recurrent classifiers flag outliers outside the trained classes [13]. Onboard learning assumes correct execution [14]. Duration and spectral catalogs supply the morphology the novel symbol is trying to see [2], [4], [11]. No row of Table 1 puts known, novel, sensor, and compute in one dictionary whose residuals cannot see the wrong cause.

III. MATERIALS AND METHODS

A. PROBLEM

Let $X \in \mathbb{R}^{D\times T}$ be background-subtracted counts from $D = 8$ detectors and $T = 64$ time bins. A linear head f_W maps a morphology feature vector $\phi(X)$ to three known classes, GRB, SGR, and solar flare. TGF is held out. A single cause c is drawn from

$$\mathcal{C} = \{\texttt{known-clean}, \texttt{novel-clean}, \texttt{sensor}, \texttt{compute}\}.$$

The operational map is accept, preserve, isolate-channel, or recompute.

The required decision $\hat{c}(X, W)$ is uniquely determined, under a single-fault assumption, by a binary signature that does not use a learned fusion of OOD scores. Compound faults are not forced into $\mathcal{C}$.

B. PHYSICS RESIDUAL, BLIND TO THE WEIGHTS

A genuine point source lights several NaI-like detectors with a shared time profile scaled by a cosine angular response [1]. Let v be the median light curve of detectors whose total counts exceed a brightness gate, and let r_{phys} be one minus the leave-one-out cosine of the worst detector against v. Dropping that detector yields a gain U_{det}. A localized spike raises r_{phys} and U_{det}. A TGF that is still a point source does not. The median, rather than a principal component, is deliberate: a single loud spike must not become the template.

C. MORPHOLOGY RESIDUAL, BLIND TO A CORRUPT HEAD

The vector $\phi(X)$ concatenates duration, peak time, full width at half maximum, variability, hardness, log counts, concentration, and smoothness of the shared profile, not of $f_W(X)$. Duration and width follow the long-standing split of bursts by their time profile [11]. Hardness uses the soft and hard counts after the worst detector has been dropped when the physics residual is already large. The residual r_{morph} is Euclidean distance to the nearest known-class centroid in that feature space, using centroids of the clean head. A flipped live weight cannot move r_{morph} except through ϕ, which does not read W.

D. ARITHMETIC RESIDUAL, BLIND TO THE SKY

Let W be stored at compile time with row-sum checksum $s^\top = 1^\top W$ [46], [47]. On a live inference the normalized residual is

$$r_{\mathrm{exec}} = \frac{|1^\top(Wx) - s^\top x|}{|1^\top(Wx)| + |s^\top x| + 1}. \tag{1}$$

In exact arithmetic a new class cannot fire r_{exec}, because $1^\top(Wx) = s^\top x$ for every input x. A weight bit-flip can. In finite precision both sums carry rounding noise that grows with the magnitude of x, so an input far from the calibration range can raise r_{exec} without any fault [50].

Under the standard model $\mathrm{fl}(a \circ b) = (a \circ b)(1 + \delta)$ with $|\delta| \le u$, the standardized input x is shared by the live product and the stored-checksum product, so its own rounding cancels. The fault-free absolute discrepancy then satisfies

$$|1^\top(Wx) - s^\top x| \le B(x) = \gamma_N(u)\Big(\sum_j \big(a_j + |s_j|\big)|x_j| + \sum_i |b_i| + \Big|\sum_i b_i\Big|\Big) + \eta, \tag{2}$$

where $a_j = \sum_i |W_{ij}|$, $\gamma_N = Nu/(1 - Nu)$, u is the unit roundoff, $N = 20$ counts the longest chain of rounded operations plus a margin, and η is an underflow term. The

TABLE 1. Comparison of closely related methods. "Part" means the column is touched only inside one undifferentiated score.

Method	OOD	Compute	Sensor	Novel kept	Physics	Isolability
MSP, energy, ODIN [15]–[17]	Yes	No	No	No	No	No
Mahalanobis, G-ODIN, ViM [18]–[20]	Yes	No	No	No	No	No
OpenMax, Scheirer, Vaze [21], [22], [24]	Yes	No	No	Part.	No	No
Gavarini [27]	Yes	Part.	No	No	No	No
ProGIP [30]	Yes	Yes	No	Part.	No	No
Dr. DNA [32]	No	Yes	No	No	No	No
Geissler, FACER [33], [34]	No	Part.	No	No	No	No
BLINK [35]	No	Yes	No	No	No	No
FT-CNN, Huang–Abraham [46], [47]	No	Yes	No	No	No	No
Ares, Li and colleagues [36], [37]	No	Yes	No	No	No	No
Dixit and colleagues [39]	No	Yes	No	No	No	No
Gertler–Singer, Isermann [42], [44]	No	No	Yes	No	Yes	Yes
John and colleagues [13]	Part.	No	No	Part.	No	No
Perera and colleagues [12]	No	No	Part.	No	Part.	No
This work	Yes	Yes	Yes	Yes	Yes	Yes

bounded rule fires when the absolute discrepancy exceeds $B(x)$. The vectors a, s, and $\sum_i |b_i|$ are stored with s and are not recomputed from the live weights. $N = 20$ was fixed before the bounded rule was evaluated.

E. SIGNATURE AND LOOKUP

Thresholds are set on clean calibration residuals only. No injected fault is used to set any threshold. The physics threshold is the 96th percentile of clean residuals, times 1.25, and clipped to $[0.38, 0.62]$. The morphology threshold is the 95th percentile times 1.25. The calibrated checksum threshold is the 99.5th percentile of clean r_{exec}, times 1.25, and floored at the unit roundoff of the inference arithmetic. The drop threshold is the constant 0.28. The signature is

$$\begin{aligned} S_{\text{phys}} &= [r_{\text{phys}} > t_{\text{phys}}], \\ S_{\text{morph}} &= [r_{\text{morph}} > t_{\text{morph}}], \\ U_{\text{det}} &= S_{\text{phys}} \wedge (U_{\text{det}}^{\text{raw}} > t_{\text{drop}}), \\ U_{\text{arith}} &= [r_{\text{exec}} > t_{\text{exec}}]. \end{aligned}$$

Table 2 is the dictionary. Figure 1 is the same lookup as a diagram. The live rule is

$$\hat{c} = \begin{cases} \texttt{compute} & \text{if } U_{\text{arith}}, \\ \texttt{sensor} & \text{else if } U_{\text{det}} \vee S_{\text{phys}}, \\ \texttt{novel-clean} & \text{else if } S_{\text{morph}}, \\ \texttt{known-clean} & \text{otherwise.} \end{cases} \tag{3}$$

Checksum first is deliberate: a corrupt head must not be allowed to mint a discovery.

Three optional bits are not part of (3). A short-window spike score uses the worst detector's two-bin excess over the median of the others, scaled by the square root of that median, with the same 96th-percentile rule and drop fraction 0.28. A Poisson likelihood ratio on one detector uses that same gate and is reported in Section VI-D. If the checksum and the sensor bit are both true, (3) still says compute. An alternative label "multiple" does not change any single-fault row.

TABLE 2. Single-fault isolability dictionary. A dot means the bit is not used for that row.

Cause	S_{phys}	S_{morph}	U_{det}	U_{arith}	Response
known-clean	0	0	0	0	accept
novel-clean	0	1	0	0	preserve
sensor	1	·	1	0	isolate channel
compute	·	·	0	1	recompute

F. SIMULATOR

Each trial draws a sky position, a family template (fast-rise exponential-decay for GRB and SGR, a wide Gaussian for a solar flare, a millisecond TGF), Poisson NaI-like counts with a cosine response, and optional faults. Sensor faults add a soft spike on a dim detector. Compute faults flip one IEEE-754 bit of a linear-head weight, mild bits 18–23 or critical bits 28–30, and revert afterward [36], [38], [51]. The head is a standardized linear classifier on $\phi(X)$, not the convolutional-recurrent network of John and colleagues [13]. Inference is emulated in binary32. Binary64 and binary16 are sensitivity checks. The physics residual never needs the head.

Class counts in Table 3 are from the HEASARC `fermigtrig` table on 17 September 2026. They justify holding TGF out as a frequent, morphologically distinct family [5], [6], and they record that the charged-particle triggers are not used as light curves. The simulator's single-detector spike only loosely mimics those particle events.

G. BASELINES

Energy and maximum-softmax rejection follow the open-set-score-as-fault-detector idea [15], [16], [27]: an unusual score

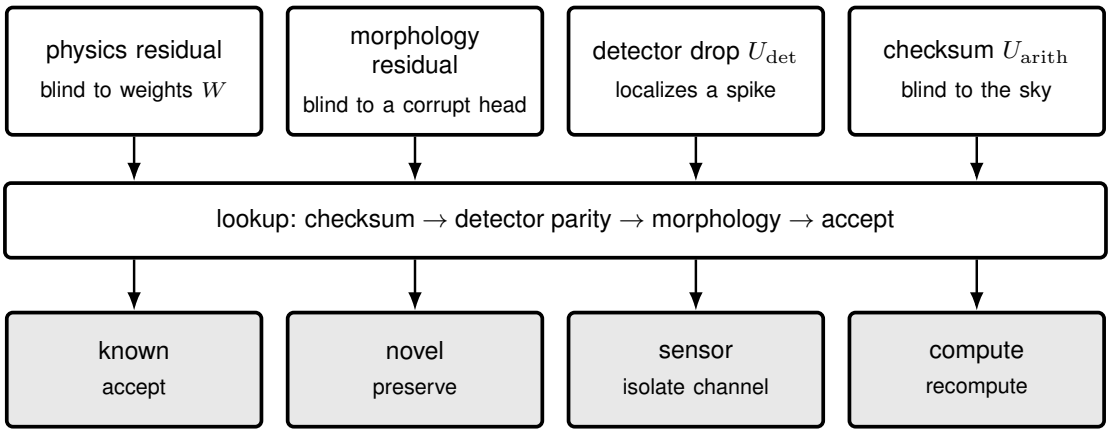


FIGURE 1. ARIA residuals and lookup. Each residual is forbidden from seeing at least one cause. The decision is a table, not a fused classifier.

TABLE 3. Public Fermi-GBM trigger types used as class priors (HEASARC `fermigtrig`, 17 September 2026).

Type	Count	Role in this paper
GRB	4375	known class
SFLARE	2800	known class
SGR	684	known class
TGF	1658	held-out novel class
LOCLPAR+DISTPAR	2167	particle events, not used as data
All triggers	12981	

is an undifferentiated anomaly. The range check follows ProGIP's magnitude idea [30]: an extreme logit magnitude is compute, a leftover unusual score is novel, and anything else is known. The range check has no sensor symbol.

H. PROTOCOLS

The headline simulator uses training seeds $\{7, 19, 31, 43, 61\}$, evaluation seed 123, and 36 trials in each cell of the three known classes clean, TGF clean, sensor on GRB and on SGR, and mild and critical compute on GRB. That is 288 four-way trials per seed. Thresholds use a disjoint clean calibration stream. Ablations reuse the same signatures with subsets of (3). Compound trials inject a spike and a bit-flip together ($n = 80$ per training seed in the signature study; the headline count uses one trained head and $n = 80$).

The headline real-data protocol trains and scores known classes on the same cubes. TGF is never in training. Sensor spikes and critical bit-flips are injected on the first 40 known cubes in file order (180 trials): every GRB in the slice, 15 SGRs, and no solar flare. The same pipeline is also reported on every known cube. Mild flips, bits 18–23, use those 40 cubes. Five-fold cross-validation, repeated five times, and a stratified half-split in both directions share no trigger name across a fold. They are still splits of one slice of 100 triggers, not a second catalog draw.

IV. RESULTS

TABLE 4. Four-way accuracy and per-cause isolation in the simulator (percent). Mean ± standard deviation over five training seeds.

Metric	ARIA	Range check
Four-way accuracy	90.0 ± 0.6	43.1 ± 6.0
Novel → novel	95.0 ± 3.2	–
Novel → compute	5.0 ± 3.2	–
Compute → compute	100.0 ± 0.0	–
Sensor → sensor	77.5 ± 0.6	0 (no sensor symbol)

TABLE 5. Per-seed simulator results (percent), evaluation seed 123. ARIA and Range are four-way accuracy. Nov., Comp., and Sens. are ARIA isolation of novel, compute, and sensor trials. E/MSP is the share of compute trials flagged by energy or maximum-softmax rejection.

Seed	ARIA	Range	Nov.	Comp.	Sens.	E/MSP
7	90.6	49.7	97.2	100.0	77.8	15.3
19	90.6	35.8	100.0	100.0	77.8	20.8
31	89.2	40.3	91.7	100.0	76.4	25.0
43	89.6	38.9	94.4	100.0	77.8	15.3
61	89.9	50.7	91.7	100.0	77.8	8.3

A. FOUR-WAY ATTRIBUTION

Table 4 reports mean and standard deviation over the five seeds. ARIA reaches $90.0 \pm 0.6\%$ four-way accuracy. The range check reaches $43.1 \pm 6.0\%$. Held-out TGF is isolated as novel in $95.0 \pm 3.2\%$ of trials and is named compute in $5.0 \pm 3.2\%$. Compute isolation is $100.0 \pm 0.0\%$. Sensor isolation is $77.5 \pm 0.6\%$: the physics residual misses some dim-detector spikes, and those trials fall through to known. That miss rate is reported rather than retuned until the number is round.

Table 5 and Fig. 2 give the per-seed values. ARIA moves by at most 1.4 points across seeds. The range check moves from 35.8% to 50.7%.

A representative confusion matrix, training seed 7, is Table 6 and Fig. 3. Energy and maximum-softmax rejection flag 100% of novel trials as an anomaly and between 8.3% and 25.0% of compute trials depending on the seed (mean 16.9%). The same rejector fires on discovery and on some bit-flips, and it cannot name either.

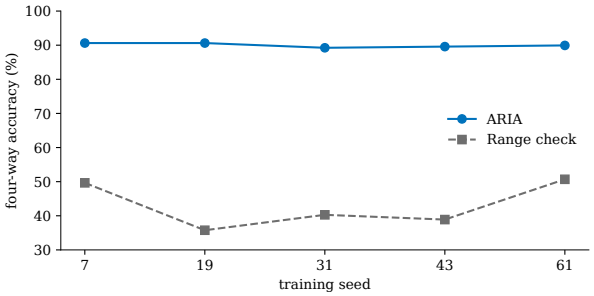


FIGURE 2. Four-way accuracy by training seed, from Table 5. ARIA stays near 90% on every seed. The range check varies by about 15 points.

TABLE 6. ARIA confusion counts for training seed 7 in the simulator (36 trials in the novel row and in each compute half; 108 known-clean trials; 72 sensor trials).

True \ predicted	known	novel	sensor	compute
known	98	0	10	0
novel	0	35	0	1
sensor	15	1	56	0
compute	0	0	0	72

B. CHECKSUM CALIBRATION, PRECISION, AND THE ROUNDING BOUND

The checksum threshold is set on clean residuals only. With that rule, compute isolation is 100% in binary32 and binary64, but the held-out TGF is named compute in $5.0 \pm 3.2\%$ of binary32 trials. A TGF lies far from the calibration range, so its rounding noise can exceed a threshold calibrated on known classes. The checksum is blind to the sky in exact arithmetic, and not exactly so in finite precision.

Table 7 compares the calibrated threshold with the bound (2) on the same trials. At binary32 the bounded rule names no TGF a compute fault ($0.0 \pm 0.0\%$), keeps compute isolation at $100.0 \pm 0.0\%$, including mild flips, and reaches $90.8 \pm 0.2\%$ four-way accuracy against $90.0 \pm 0.6\%$ for the calibrated rule. Figure 4 shows why. In the calibrated panel a share of the TGF trials sits above the dotted threshold. In the bounded panel every fault-free trial, TGF included, is below 1, and every flip in the protocol is above it. Figure 6 shows that at binary32 both statistics rank every compute trial above every TGF trial, so the false-compute rate of the calibrated rule comes from where its threshold sits, not from the residual being unable to rank the two causes.

The guarantee has a price at binary16. Compute isolation falls to $46.1 \pm 1.0\%$ for the bound against $84.2 \pm 5.0\%$ for the calibrated rule, and mild flips are caught in $15.6 \pm 1.4\%$ of trials against $76.1 \pm 6.0\%$ (Fig. 7). Figure 5 breaks detection down by the flipped bit. The lowest mantissa bits are missed by every variant, because they move a weight by less than the arithmetic's own noise. At binary32 the calibrated rule reaches high detection from bit 8 and the bounded rule from bit 12. The bounded rule trades sensitivity for a guarantee of no false alarm on fault-free inputs.

As a soundness check, 40,000 fault-free random inputs per precision were drawn up to 1000 standard deviations from

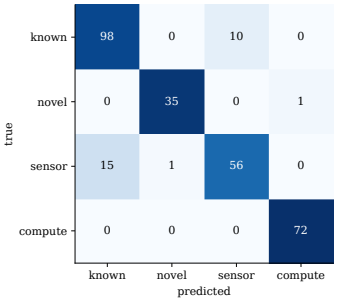


FIGURE 3. ARIA confusion counts for training seed 7, the same counts as Table 6. Shading is the row-normalized rate. The main errors are missed sensor spikes, clean events called sensor, and one held-out TGF called compute.

the training mean. The largest ratio of residual to bound was 0.15, and none exceeded 1. This checks the bound on this head. It is not a proof for other heads, and a fault in the stored vectors themselves is outside the model.

On real CTIME the same comparison is Table 8. At binary32 the bounded rule keeps compute isolation at 100% and names no TGF compute at any slice size, while the calibrated rule names 8.3% of TGFs as compute at 12 triggers per class. Sensor isolation at eight triggers per class is 70.8% for the bound against 58.3% for the calibrated rule, because some spiked cubes trip the calibrated checksum and are therefore labeled compute before the sensor bit is read. At the 25-cube slice the two rules agree. Novel isolation stays at 0% in every row. That limit comes from the CTIME bins, not from the checksum.

C. ABLATIONS

Table 9 and Fig. 8 remove signature bits. A checksum alone cannot name novelty or sensors ($62.4 \pm 0.2\%$). Physics alone names some sensors and defaults everyone else to known ($53.5 \pm 0.0\%$). Morphology alone is an open-set detector ($49.5 \pm 0.5\%$). Removing only the checksum drops accuracy to $65.8 \pm 0.2\%$, because compute trials fall into known or novel. Full ARIA is $90.0 \pm 0.6\%$. Each residual covers a cause the others cannot see.

D. SUMMARY AGAINST THE BASELINES

Table 10 puts ARIA next to the two baselines on the simulator and on real CTIME. Only ARIA names a cause for novelty, compute faults, and sensors. On novelty it pays a small false-compute rate in the simulator under the calibrated checksum. Energy and maximum-softmax rejection never name a cause.

E. COMPOUND FAULTS

When a spike and a bit-flip occur together, U_{arith} still fires, so the headline lookup names compute on 100.0% of 80 compound trials (Table 11). Across five training seeds, both the checksum bit and the sensor bit are true on $79.2 \pm 3.1\%$ of compound trials, and the checksum alone is true on $20.7 \pm 3.1\%$ (Fig. 9). The physics bit never fires by itself. The optional two-fault rule therefore returns "multiple" on $79.2 \pm 3.1\%$ of these trials and still returns compute on the

TABLE 7. Calibrated and rounding-error-bounded checksum in the simulator (percent, mean ± standard deviation over five seeds).

Checksum	Prec.	Acc.	Nov.→comp.	Comp. (all)	Comp. (mild)	Known→comp.
calibrated	fp64	90.1 ± 0.5	3.9 ± 4.2	100.0 ± 0.0	100.0 ± 0.0	0.4 ± 0.5
calibrated	fp32	90.0 ± 0.6	5.0 ± 3.2	100.0 ± 0.0	100.0 ± 0.0	0.4 ± 0.5
calibrated	fp16	86.3 ± 1.2	4.4 ± 6.2	84.2 ± 5.0	76.1 ± 6.0	0.2 ± 0.4
bounded	fp64	90.8 ± 0.2	0.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
bounded	fp32	90.8 ± 0.2	0.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	0.0 ± 0.0
bounded	fp16	77.4 ± 0.3	0.0 ± 0.0	46.1 ± 1.0	15.6 ± 1.4	0.0 ± 0.0

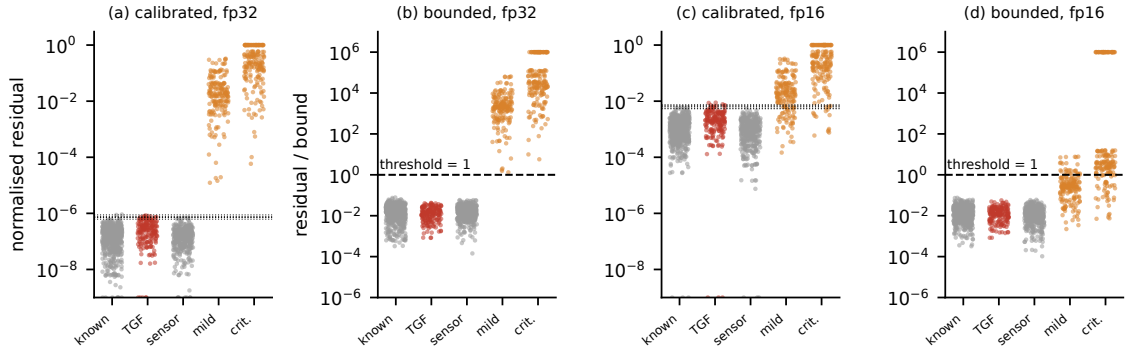


FIGURE 4. Checksum statistic by true cause, five seeds pooled (counts in Table 7). Panels (a) and (c) are the calibrated residual. Panels (b) and (d) divide it by the bound (2). (a) and (b) are binary32. (c) and (d) are binary16.

TABLE 8. Calibrated and bounded checksum on real CTIME (%, mean over five training seeds; the across-seed spread is 0.0 in every cell, as explained in Section V). Triggers per class; compute faults are critical flips only. Nov.→comp.: held-out TGF named compute.

Checksum	Prec.	n/cl.	Acc.	Comp.	Sens.	Nov.→comp.
calibrated	fp32	8	76.3	100.0	58.3	0.0
calibrated	fp32	12	76.7	100.0	61.1	8.3
calibrated	fp32	25	74.4	100.0	60.0	0.0
calibrated	fp16	8	73.8	100.0	54.2	12.5
calibrated	fp16	12	77.5	100.0	63.9	0.0
calibrated	fp16	25	74.4	100.0	60.0	0.0
bounded	fp32	8	80.0	100.0	70.8	0.0
bounded	fp32	12	77.5	100.0	63.9	0.0
bounded	fp32	25	74.4	100.0	60.0	0.0
bounded	fp16	8	73.8	79.2	70.8	0.0
bounded	fp16	12	75.0	91.7	63.9	0.0
bounded	fp16	25	73.3	95.0	60.0	0.0

rest, where the spike left no physics signature. The dictionary can refuse a single label when both residuals fire. It cannot recover a sensor fault that the physics residual missed. Two causes remain unisolable as a pair of named faults.

V. PUBLIC CTIME MEASUREMENTS

A. HEADLINE SLICE

The download keeps 25 triggers of each of GRB, SGR, solar flare, and TGF, binned to 8×64 light cubes (100 cubes). CTIME bins are coarse, nominally 64 ms and 256 ms in the background regime. A TGF is sub-millisecond [5], [6].

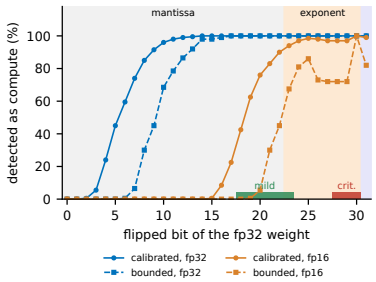


FIGURE 5. Share of single bit-flips detected as compute, against the flipped bit of the binary32 weight (200 flips per bit, five seeds, fault-free known-class events). Circles are the calibrated rule. Squares are the bounded rule. The shaded bands mark mantissa, exponent, and sign. The bars at the bottom mark the mild range (bits 18–23) and the critical range (bits 28–30) used in the main protocol.

TABLE 9. Ablation of the lookup. Four-way accuracy, percent, mean ± standard deviation over five seeds.

Lookup	Accuracy
Full ARIA, equation (3)	90.0 ± 0.6
Checksum only	62.4 ± 0.2
Physics only	53.5 ± 0.0
Morphology only	49.5 ± 0.5
No checksum	65.8 ± 0.2

The linear head is trained on all known-class cubes in the slice. Known-clean evaluation reuses those cubes, so four-way accuracy is optimistic on the known-clean rows. TGF is held out entirely. Real-data compute trials use critical flips only in the headline row. The run uses the same five training

TABLE 10. Headline results (percent, mean ± standard deviation over five training seeds). A dash means that output is not produced. Flag novel and flag compute are undifferentiated anomaly marks.

Setting	Method	Four-way	Novel	Compute	Sensor	Nov.→comp.	Flag novel	Flag compute
Simulator	ARIA	90.0 ± 0.6	95.0 ± 3.2	100.0 ± 0.0	77.5 ± 0.6	5.0 ± 3.2	–	–
	Range check	43.1 ± 6.0	–	–	0	–	–	–
	Energy / MSP	–	–	–	–	–	100.0 ± 0.0	16.9 ± 5.7
Real CTIME	ARIA	74.4 ± 0.0	0.0 ± 0.0	100.0 ± 0.0	60.0 ± 0.0	0.0 ± 0.0	–	–
	Range check	43.3 ± 0.0	–	–	0	–	–	–
	Energy / MSP	–	–	–	–	–	0.0 ± 0.0	27.5 ± 0.0

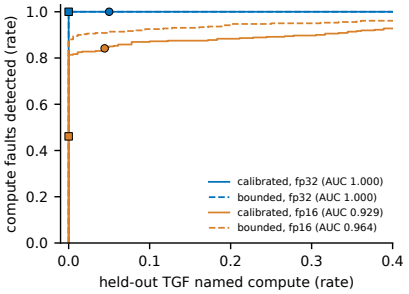


FIGURE 6. Compute faults detected, against held-out TGFs named compute, as the checksum threshold is swept (five seeds pooled). The marker is the operating point actually used: the per-seed calibrated threshold, or the bound threshold 1.

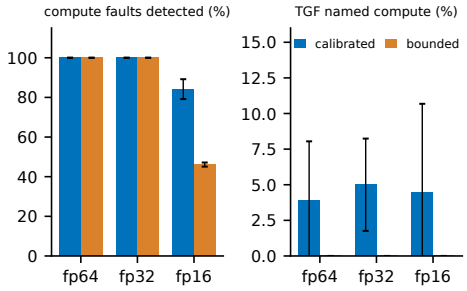


FIGURE 7. Calibrated and bounded checksums at three inference precisions (mean ± standard deviation over five seeds; values in Table 7). The bounded rule never names a TGF compute in the simulator. It loses compute sensitivity at binary16.

seeds and evaluation seed 123, with 180 trials per seed.

Table 12 and Fig. 10 compare that run with the simulator. Compute isolation on critical flips is 100.0%. Held-out TGF is named compute in 0.0% of trials at this slice size. Four-way accuracy is 74.4% and sensor isolation is 60.0%. Novel isolation is 0.0%. On these bins a TGF is not separable from a short GRB, so held-out TGFs fall through to known. Table 13 shows the calls at training seed 7: all 25 TGF cubes are called known, and 29 of 40 sensor injections are caught. This is a negative result about the discovery symbol, not a tuning failure. Section V-C shows that a lower morphology threshold does not fix it.

The across-seed spread on this slice is zero for every headline metric, because the same cubes train every seed. Slice size, cross-validation, and the half-split are the checks on that axis.

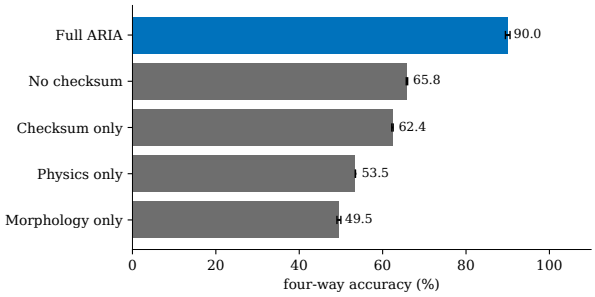


FIGURE 8. Ablation of the lookup. Values are in Table 9. Every residual alone is far below the full dictionary.

TABLE 11. Headline lookup on 80 compound sensor-plus-compute trials from one trained head. The five-seed signature split is in Fig. 9.

	known	novel	sensor	compute
Trials named	0	0	0	80

B. SLICE SIZE

Figure 11 and Table 14 repeat the calibrated binary32 pipeline at three slice sizes. The smaller slices are prefixes of the 25-cube families. Four-way accuracy is 76.3%, 76.7%, and 74.4%. It is not monotone. Compute isolation is 100% at every size. Novel isolation is 0% at every size. The held-out TGF is named compute in 0%, 8.3%, and 0% of trials, so the false-compute rate on novelty is small, not constant, and not a structural zero.

C. WHY THE NOVEL SYMBOL IS EMPTY

Figure 12 and Table 15 plot the morphology residual on the clean cubes. The mean distance of TGF to the nearest known centroid is smaller than the mean distance of GRB. No TGF exceeds the shipped threshold. Four solar flares do, which is why a known class, not the held-out class, supplies the morphology false alarms. In the duration–width plane the four families overlap. TGF does not occupy a region the three trained classes leave empty, once the light curve has been averaged into CTIME bins.

Table 16 and Fig. 13 lower and raise that threshold. The shipped margin is 1.25 on the clean 95th percentile. At margin 0.5, novel isolation rises only to 12%, while 41.3% of known triggers are called novel and four-way accuracy falls to 60%. There is no operating point on this slice at which TGF is recovered and known bursts are still accepted. The

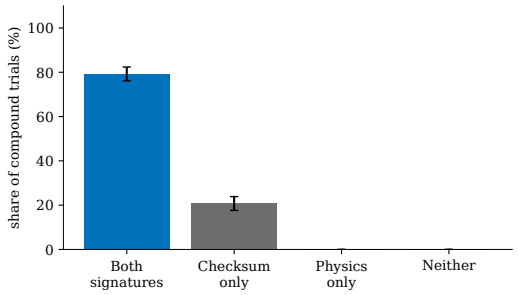


FIGURE 9. Which residual bits fire on compound trials (mean ± standard deviation over five training seeds, 80 trials each). "Both signatures" is the share the optional rule labels multiple. The headline lookup labels every one of these trials compute.

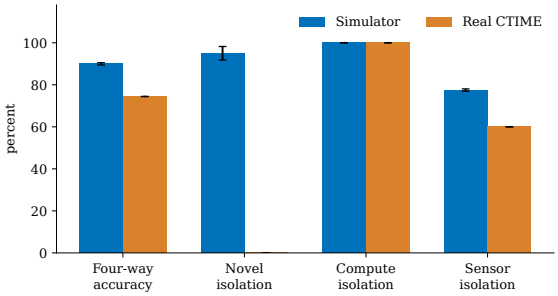


FIGURE 10. ARIA on the simulator and on real CTIME, from Table 12. Compute isolation is high in both settings. The real compute trials are critical flips only. Sensor isolation and four-way accuracy drop. Novel isolation is zero on CTIME.

TABLE 12. Public GBM CTIME, 25 triggers per class, software fault injection (percent). Mean ± standard deviation over 5 training seeds at evaluation seed 123. The simulator column is Table 4.

Metric	Real CTIME	Simulator
Four-way accuracy	74.4 ± 0.0	90.0 ± 0.6
Range-check accuracy	43.3 ± 0.0	43.1 ± 6.0
Novel → novel	0.0 ± 0.0	95.0 ± 3.2
Novel → compute	0.0 ± 0.0	5.0 ± 3.2
Compute → compute	100.0 ± 0.0	100.0 ± 0.0
Sensor → sensor	60.0 ± 0.0	77.5 ± 0.6

TABLE 13. ARIA calls on the public CTIME slice at training seed 7 and evaluation seed 123, using the published injection budget (the first 40 known cubes). All 25 held-out TGF cubes are called known.

True \ predicted	known	novel	sensor	compute
known	70	2	2	1
novel	25	0	0	0
sensor	16	0	24	0
compute	0	0	0	40

discovery symbol needs finer time bins. It does not need a lower threshold.

VI. CHECKS THAT WERE NOT IN THE HEADLINE TABLE

A. OUT-OF-SAMPLE CUBES, AND THE INJECTION BUDGET

The headline real-data accuracy trains and scores known classes on the same cubes, and it injects faults only into the first 40 known cubes. Table 17 shows what that budget is. Every GRB is injected, 15 of 25 SGRs are injected, and no solar flare is injected. When every known cube is injected, sensor isolation rises from 60.0% to $69.3 \pm 0.0\%$ and four-way accuracy from 74.4% to $78.8 \pm 0.0\%$. GRB spikes are caught 56% of the time. Solar-flare spikes are caught 80% of the time, and the SGR rate moves from 66.7% on the 15 cubes inside the prefix to 72% once all 25 are included. The 60% figure is the GRB-heavy budget, not a property of every family.

Table 18 and Fig. 14 score cubes the head was not trained on. Five-fold cross-validation, repeated five times, gives $77.2 \pm 2.7\%$ accuracy, $100.0 \pm 0.0\%$ compute isolation, $65.3 \pm 7.5\%$ sensor isolation, and $0.0 \pm 0.0\%$ of TGFs named compute. Novel isolation remains $0.0 \pm 0.0\%$. The half-split agrees in both directions: accuracy $77.7 \pm 0.0\%$ and $80.0 \pm 0.0\%$, sensor isolation $66.7 \pm 0.0\%$ and $72.2 \pm 0.0\%$, compute isolation 100%, and no TGF named compute. The in-sample headline is not an outlier relative to these splits of one slice. The seed-to-seed standard deviation of zero on the headline row was the wrong uncertainty. The fold-to-fold sensor spread, ±7.5 points, is the uncertainty this slice can support. A second catalog draw is reported below, and it moves sensor isolation by much more than the folds do. Mild bit-flips, bits 18–23, are isolated as compute on $100.0 \pm 0.0\%$ of the published injection budget, the same rate as critical flips, under the calibrated binary32 rule. The flips that stay invisible are the low mantissa bits in Fig. 5, not this mild band.

B. SHORT-WINDOW SPIKE SCORE

The physics residual is a full-curve cosine. A two-bin spike is a small fraction of a 64-bin inner product, which is a plausible reason sensor isolation is the weak cell. The short-window score in Section 3 was built to test that reason. Its threshold uses clean cubes only.

Table 19 and Fig. 15 are the result. On the simulator, adding the bit raises sensor isolation from $77.5 \pm 0.6\%$ to $86.7 \pm 0.7\%$ and four-way accuracy from the headline to $92.2 \pm 0.6\%$, while known events called sensor stay near $9.3 \pm 0.0\%$ and $9.4 \pm 0.4\%$. On the published CTIME budget the same bit leaves sensor isolation at $60.0 \pm 0.0\%$ and accuracy at $74.4 \pm 0.0\%$. Novel events called sensor stay at $0.0 \pm 0.0\%$. The obvious short-window correction repairs the simulator and does not repair the public cubes. Section VI-D replaces that Gaussian excess with a Poisson likelihood ratio whose amplitude is maximized. The response-folded template bank of Perera and colleagues [12] is a different measurement, and it is not the one in that section.

C. A SECOND CATALOG DRAW

The half-split reshuffles one download. Table 20 is a second download: 25 triggers in each of the four families, no name in common with the published slice, binned by the same CTIME procedure. Training on draw 1 and scoring draw 2 gives 70.6% four-way accuracy, 100% compute isolation, 65% sensor isolation, and no TGF called novel or compute.

TABLE 14. Calibrated fp32 ARIA on public CTIME as the slice grows (percent, mean over five training seeds). The 8- and 12-cube rows are prefixes of the 25-cube slice.

Per class	Acc.	Compute	Sensor	Novel→comp.	Novel
8	76.3	100.0	58.3	0.0	0.0
12	76.7	100.0	61.1	8.3	0.0
25	74.4	100.0	60.0	0.0	0.0

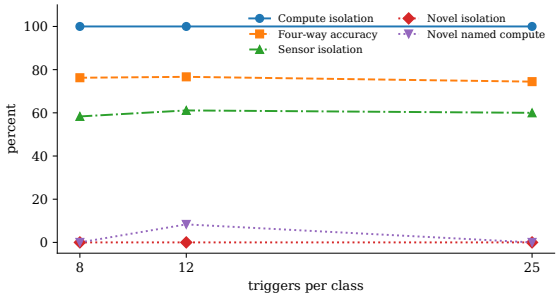


FIGURE 11. Real CTIME metrics against slice size, calibrated binary32, from Table 14. Compute isolation stays at 100%. Novel isolation stays at 0%. The false-compute rate on TGF is not zero at every size.

TABLE 15. Morphology distance to the nearest known-class centroid on clean public CTIME cubes (training seed 7). The shipped threshold is 3.98. TGF is closer to the known centroids than GRB is. Above: cubes whose distance exceeds that threshold.

Family	n	Mean	Min	Max	Above
GRB	25	1.18	0.48	2.87	0
SGR	25	1.36	0.39	3.18	0
SFLARE	25	2.65	0.76	9.59	4
TGF	25	1.08	0.57	2.47	0

TABLE 16. Morphology-threshold sweep on the public CTIME slice (training seed 7, published injection budget). The margin multiplies the 95th percentile of clean known-class morphology distances. The shipped rule uses margin 1.25. Lowering it does not recover TGF as novel without calling a large share of known triggers novel.

Margin	Acc.	Novel	Known→novel	Novel→known
0.5	60.0	12.0	41.3	88.0
0.75	72.2	4.0	9.3	96.0
1	74.4	0.0	2.7	100.0
1.25	74.4	0.0	2.7	100.0
1.5	75.0	0.0	1.3	100.0
2	75.6	0.0	0.0	100.0
3	75.6	0.0	0.0	100.0
4	75.6	0.0	0.0	100.0

TABLE 17. Sensor isolation by trigger family on public CTIME (percent, pooled over five training seeds). The count in parentheses is the number of injected cubes of that family. The published budget never injects a spike into a solar flare.

Budget	GRB	SGR	SFLARE
First 40 known cubes	56.0 (25)	66.7 (15)	–
Every known cube	56.0 (25)	72.0 (25)	80.0 (25)

The other direction gives 62.8% accuracy and 7.5% sensor isolation, with compute isolation still 100% and the novel symbol still empty. Draw 2 scored on itself is already at 25% sensor isolation.

The mechanism is the clip on the physics threshold. On draw 1 the clean residuals sit low, so the threshold takes its floor, 0.38. On draw 2 they sit high, so the threshold takes its ceiling, 0.62. Injected spikes on either draw have a median physics residual near 0.4. They clear 0.38 about as often as the headline 60%, and they mostly miss 0.62. Compute isolation and the empty novel symbol are the results that survive a new draw. The sensor rate does not.

Removing the clip, and refitting the same percentile, widens the disagreement. Training on draw 1 and scoring draw 2 then isolates 87.5% of spikes and names 9.3% of clean known cubes a sensor fault. The other direction isolates 5.0%. The floor and the ceiling were pulling the two thresholds toward each other. Table 20 keeps the clip.

D. POISSON LIKELIHOOD RATIO ON ONE DETECTOR

The short-window score is a Gaussian excess against the other detectors. Perera and colleagues search the same instrument with a Poisson matched filter, folded through the detector response and a bank of spectral templates [12]. For a flat template and similar backgrounds their fixed-amplitude statistic reduces to that Gaussian excess. These cubes do not contain a response matrix. A glitch amplitude is not a GRB detection limit, so the bit measured here maximizes the amplitude on the counts that are already stored.

For each detector, and for the soft channel and the hard channel separately, the background is the upper median of the first eight bins and the last eight bins. That is the same edge used to subtract background elsewhere, and it is at least 0.5 counts so the logarithm is defined. On a bin with value d and background b, the likelihood ratio for a non-negative excess is $d\log(d/b) - (d - b)$ when $d > b$, and zero otherwise. The detector score is the maximum, over start time, of that ratio on two adjacent soft bins plus two adjacent hard bins. The event score is the largest detector. The bit fires when the score exceeds the 96th percentile of clean known cubes on the training draw, times 1.25, and dropping the winning detector reduces the score by more than 0.28. The threshold is fit on the training draw only. If the lookup already says compute, the call stays compute. Otherwise a firing bit is read as sensor. The injected spike is added only to the soft channel, which is the channel this ratio sees.

Table 21 and Fig. 16 report every protocol. The lookup column matches the rate already given for that protocol. The split between 100% and 55% is the threshold. On draw 1 it is 743.9. The median clean score is 39.4 and the median

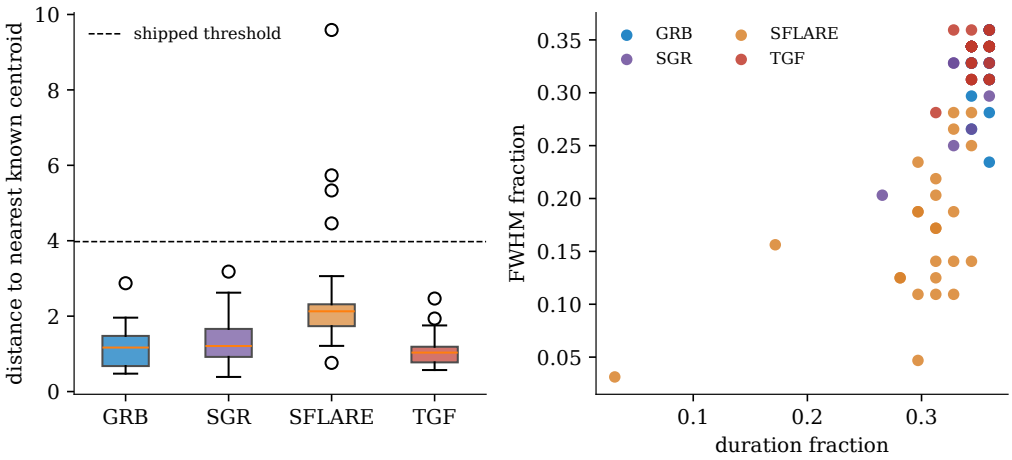


FIGURE 12. Left: morphology distance of each clean CTIME cube to the nearest known-class centroid. The dashed line is the shipped threshold. Right: duration fraction against full width at half maximum for the same cubes. TGF sits inside the GRB cloud.

TABLE 18. Public CTIME outside the headline protocol (percent). Neither split is a new catalog draw.

Protocol	Acc.	Compute	Sensor	Nov.→comp.	Novel
In-sample, first 40 known cubes	74.4 ± 0.0	100.0 ± 0.0	60.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
In-sample, every known cube	78.8 ± 0.0	100.0 ± 0.0	69.3 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Mild flips, first 40 known cubes	74.4 ± 0.0	100.0 ± 0.0	60.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
5×5 cross-validation	77.2 ± 2.7	100.0 ± 0.0	65.3 ± 7.5	0.0 ± 0.0	0.0 ± 0.0
Half-split, A→B	77.7 ± 0.0	100.0 ± 0.0	66.7 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
Half-split, B→A	80.0 ± 0.0	100.0 ± 0.0	72.2 ± 0.0	0.0 ± 0.0	0.0 ± 0.0

TABLE 19. Short-window spike score, thresholded on clean cubes only and OR-ed into the sensor bit (percent). On the simulator the bit raises sensor isolation. On the published CTIME budget it does not.

Rule	Acc.	Sensor	Known→sensor	Novel→sensor
Simulator, headline	90.0 ± 0.6	77.5 ± 0.6	9.3 ± 0.0	0.0 ± 0.0
Simulator, spike bit added	92.2 ± 0.6	86.7 ± 0.7	9.4 ± 0.4	0.0 ± 0.0
CTIME, headline	74.4 ± 0.0	60.0 ± 0.0	2.7 ± 0.0	0.0 ± 0.0
CTIME, spike bit added	74.4 ± 0.0	60.0 ± 0.0	2.7 ± 0.0	0.0 ± 0.0

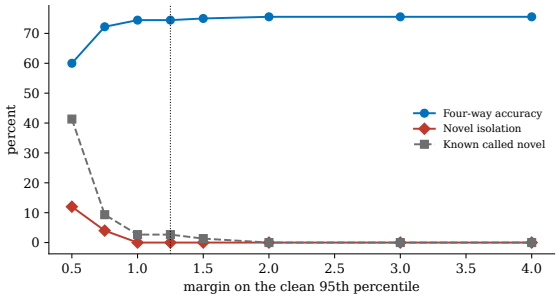


FIGURE 13. Threshold sweep from Table 16. The dotted line is the shipped margin. Novel isolation stays near zero until known triggers are being rejected in large numbers.

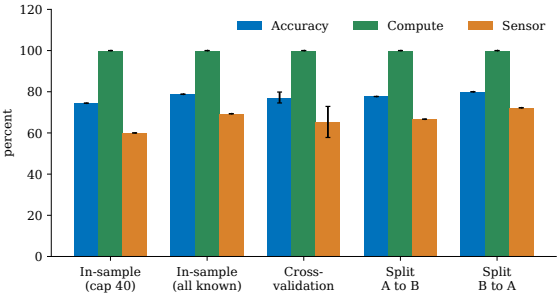


FIGURE 14. Public CTIME under the protocols of Table 18. Compute isolation stays at 100%. Accuracy stays in a band around the headline 74.4%. Sensor isolation is the quantity that moves.

spike score is 3278.6, so every injected spike clears it. On draw 2 it is 3190.3. The median clean score is 57.4. The 96th percentile is set by a few clean magnetar bursts and flares that already have a large one-detector excess, and the median spike score on that draw is 3297.5, so the threshold sits on the spikes. The bit fires on 52.5% of them. The union with

TABLE 20. Disjoint CTIME draws (percent). Draw 2 shares no trigger name with the published slice. Seed-to-seed spread is zero.

Train → score	Accuracy	Compute	Sensor	Novel	Novel→compute
Draw 1 → draw 1	74.4	100	60.0	0	0
Draw 1 → draw 2	70.6	100	65.0	0	0
Draw 2 → draw 2	67.8	100	25.0	0	0
Draw 2 → draw 1	62.8	100	7.5	0	0

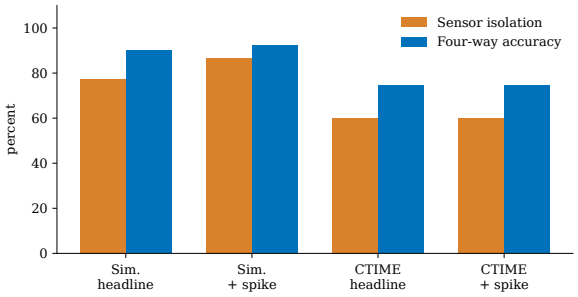


FIGURE 15. Sensor isolation and four-way accuracy with and without the short-window spike bit, from Table 19. The bit helps the simulator. It does not move the public CTIME result.

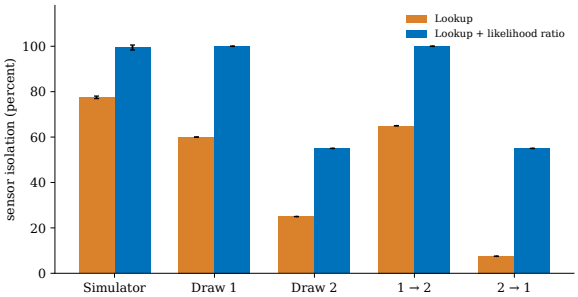


FIGURE 16. Sensor isolation from Table 21. Orange is the lookup. Blue adds the Poisson likelihood-ratio bit. The bit reaches 100% when the threshold is fit on draw 1. A threshold fit on draw 2 stops it at 55%. Error bars are the seed-to-seed standard deviation and are zero on the CTIME bars.

the lookup is the 55.0% in the table. On every spiked trial in the table, the detector with the largest likelihood ratio is the injected detector.

On the simulator the same bit raises sensor isolation from $77.5 \pm 0.6\%$ to $99.4 \pm 1.1\%$ and accuracy from $90.0 \pm 0.6\%$ to $92.5 \pm 0.7\%$. It also fires on flashes. Novel isolation falls from $95.0 \pm 3.2\%$ to $74.4 \pm 2.1\%$, and $20.6 \pm 2.2\%$ of flashes are called sensor. A point source near one detector is brighter there. This ratio does not divide by a detector response, so that brightness can clear the drop-gain gate. The median flash score is 3249.9, above the simulator threshold of 1569.9 ± 185.1. On the public cubes the same flashes do not clear the gate: the bit fires on none of the TGFs in either draw. Critical bit-flips remain compute on every trial. The bit is true on 5.0% of the draw-2 compute injections and on $1.1 \pm 0.6\%$ of the simulator compute injections, and the checksum-first rule leaves those trials as compute.

E. WHICH DETECTOR THE SPIKE LANDS ON

The sensor symbol says isolate the channel. Table 17 says how often that symbol fires. It does not say whether the worst detector is the detector that received the spike. Table 22 and

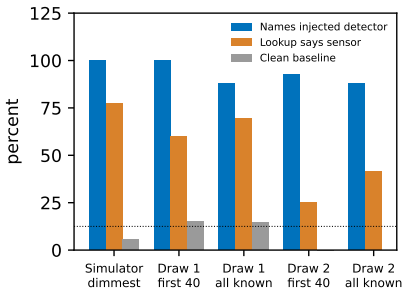


FIGURE 17. Channel naming from Table 22. Blue: the worst detector is the injected detector. Orange: the lookup says sensor. Gray: the same index on the clean cube. The dotted line is a uniform guess over eight detectors, 12.5%. Draw 2, all known cubes, has no gray bar.

Fig. 17 score that index on the same injections. The published injector puts the spike on the dimmest detector that is not the brightest. The clean baseline is that index on the same cube with no spike. A uniform guess over eight detectors would be right 12.5% of the time.

On the simulator the injected detector is the worst detector in all 72 trials per seed, including every trial the lookup calls sensor. The clean baseline is 5.6%. Drawing the detector uniformly, which the other tables do not do, names it in 97.2% of trials and in every trial called sensor. On the first 40 known cubes of draw 1 the worst detector is the injected one in every trial, and every sensor call names it. The clean baseline is 15%. Across all 75 known cubes the index matches in 88.0% of trials and in 92.3% of the trials called sensor. The misses are not spread evenly: all 25 GRB spikes match, 24 of 25 SGR spikes match, and 17 of 25 solar-flare spikes match. Draw 2 keeps the index when it drops the sensor call. On its first 40 known cubes the lookup says sensor on 25% of trials, the worst detector is still the injected one on 92.5%, and the clean baseline is 0. Of the trials called sensor, 80% name the injected detector. Across all 75 known cubes of draw 2 the match is 88.0%, and 83.9% among sensor calls.

The weak cell is the decision to say sensor. The channel named by the physics residual is usually the channel that was spiked.

F. MILLISECOND LIGHT CURVES

The CTIME novel symbol is empty because a sub-millisecond flash occupies one 64 ms bin. The same names were rebinned from time-tagged files at 1 ms, from 0.5 s before the trigger to 1.5 s after. `bn260414904` had no usable eight-detector file in that first download, leaving 25 bursts and 24 flashes. Background is the median of $[-0.5, -0.1)$ s. Distance is Euclidean in the eight morphology features. Each burst is scored against the other bursts, and the threshold is

TABLE 21. Maximized-amplitude Poisson likelihood ratio, OR-ed into the sensor call unless the lookup already says compute (percent, mean ± standard deviation over five seeds). No detector response was used.

Rule	Acc.	Sensor	Known→sensor	Novel→sensor	Novel	Compute
Simulator, lookup	90.0 ± 0.6	77.5 ± 0.6	9.3 ± 0.0	0.0 ± 0.0	95.0 ± 3.2	100.0 ± 0.0
Simulator, likelihood-ratio bit	92.5 ± 0.7	99.4 ± 1.1	10.4 ± 0.9	20.6 ± 2.2	74.4 ± 2.1	100.0 ± 0.0
Draw 1, lookup	74.4 ± 0.0	60.0 ± 0.0	2.7 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 1, likelihood-ratio bit	82.8 ± 0.0	100.0 ± 0.0	4.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 2, lookup	67.8 ± 0.0	25.0 ± 0.0	2.7 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 2, likelihood-ratio bit	73.9 ± 0.0	55.0 ± 0.0	4.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 1 → draw 2, lookup	70.6 ± 0.0	65.0 ± 0.0	8.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 1 → draw 2, likelihood-ratio bit	77.8 ± 0.0	100.0 ± 0.0	9.3 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 2 → draw 1, lookup	62.8 ± 0.0	7.5 ± 0.0	2.7 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0
Draw 2 → draw 1, likelihood-ratio bit	73.3 ± 0.0	55.0 ± 0.0	2.7 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	100.0 ± 0.0

TABLE 22. Whether the worst detector is the injected detector (percent). A uniform guess over eight detectors is 12.5%. The clean baseline uses the same index with no spike.

Setting	N	Called sensor	Names channel	Right | sensor	Clean baseline
Simulator, published injector	72	77.5 ± 0.6	100	100	5.6
Simulator, uniform detector	72	63.9	97.2	100	–
CTIME draw 1, first 40 known	40	60.0	100	100	15.0
CTIME draw 1, all known	75	69.3	88.0	92.3	14.7
CTIME draw 2, first 40 known	40	25.0	92.5	80.0	0
CTIME draw 2, all known	75	41.3	88.0	83.9	–

their 95th percentile times 1.25. Figure 18 shows the result. The threshold is 5.21. One of 25 bursts lies above it, and 23 of 24 flashes do. The flash that does not, `bn120312139`, has a full width at half maximum of 1.2 s, against about 1 ms for the flashes that separate. This is a GRB-only distance, not a lookup label.

The four-family lookup uses a later file from the same window, 25 triggers in each class, including `bn260414904`, with the same pre-trigger background and no injected fault. All five seeds agree. The physics threshold sits on its ceiling, 0.62. Median physics residuals are 0.986 (GRB), 0.974 (SGR), 1 (solar flare), and 0.954 (TGF), so the bit is on for every burst, every magnetar burst, every flash, and 23 of 25 flares. The morphology threshold is 5.85 and is true for 1 flash, 1 magnetar burst, 2 flares, and no burst. Those two flares are the only novel calls. The checksum threshold is 6.31×10^{-7} and the median checksum on flashes is 1.40×10^{-7}, so the compute symbol stays off. Table 23 names every flash a sensor fault. At 8, 16, and 50 ms the same photons, with the ordinary eight-bin background, keep that call: median physics residuals are 1 in every family at 8 and 16 ms, and 1 for bursts, magnetar bursts, and flashes at 50 ms (0.92 for flares). At 8 ms, 24 flashes are sensor and one is compute. At 16 and 50 ms all 25 flashes are sensor. At 50 ms, seven flares and two magnetar bursts fall through to known, and one flare is named compute.

TABLE 23. Lookup calls on time-tagged photons, with no injected fault. Each count is out of 25. Every training seed returns the same row. The 1 ms row subtracts a pre-trigger background. The other rows sum that file and use the first and last eight bins.

Bin	Family	Sensor	Known	Novel	Compute
1 ms	GRB	25	0	0	0
	SGR	25	0	0	0
	Solar flare	23	0	2	0
	TGF	25	0	0	0
8 ms	GRB	25	0	0	0
	SGR	25	0	0	0
	Solar flare	22	2	1	0
	TGF	24	0	0	1
16 ms	GRB	25	0	0	0
	SGR	25	0	0	0
	Solar flare	21	2	2	0
	TGF	25	0	0	0
50 ms	GRB	24	0	1	0
	SGR	23	2	0	0
	Solar flare	16	7	1	1
	TGF	25	0	0	0

G. LOOKUP AGAINST A LABELED FUSION

The lookup is a table. Table 24 asks what a learned combiner does with the same four numbers. The combiner is a class-weighted softmax on the physics residual, the morphology residual, $\log_{10}$ of the checksum, and the drop gain. The logarithm is a monotone rescaling: the raw checksum spans about eight decades, and a linear model standardized on that raw value cannot see a separation that a threshold can see. Training uses a disjoint simulator stream and includes flashes labeled novel. The lookup's thresholds are still fit on clean known-class residuals only. Both are scored on the published evaluation stream.

On single faults the fusion is the more accurate rule: $96.5 \pm 0.3\%$ four-way accuracy against $90.0 \pm 0.6\%$, sensor isolation $96.7 \pm 0.7\%$ against $77.5 \pm 0.6\%$, and no held-

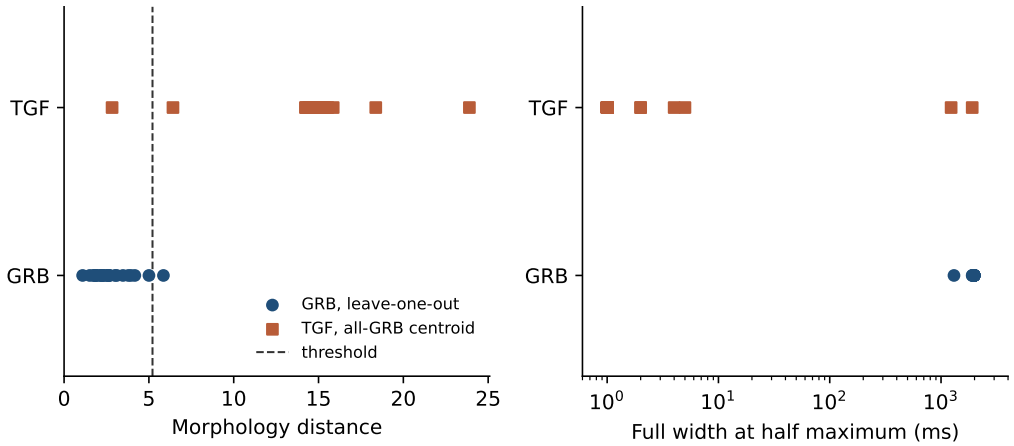


FIGURE 18. Morphology on 1 ms TTE for the published GRB and TGF names. Left: distance to the GRB centroid. Bursts are leave-one-out. Flashes use the centroid of all 25 bursts. The dashed line is the 95th percentile of the burst distances, times 1.25. Right: full width at half maximum of the background-subtracted sum, on a logarithmic axis.

TABLE 24. Lookup against a labeled fusion of the same four residuals on the simulator (percent, mean $\pm$ standard deviation over five training seeds). The fusion is a class-weighted softmax on the physics residual, the morphology residual, $\log_{10}$ of the checksum, and the drop gain. It is trained on a disjoint stream that includes flashes labeled novel. The lookup never receives those labels. Compound rows are the 80 spike-plus-bit-flip trials. The lookup column is the decision, so a compound trial whose sensor bit is also true is still counted as compute.

Metric	Lookup	Fusion
Four-way accuracy	90.0 ± 0.6	96.5 ± 0.3
Novel isolation	95.0 ± 3.2	100
Novel $\rightarrow$ compute	5.0 ± 3.2	0
Compute isolation	100	98.6
Sensor isolation	77.5 ± 0.6	96.7 ± 0.7
Compound $\rightarrow$ compute	100	27.5 ± 7.5
Compound $\rightarrow$ sensor	0	72.5 ± 7.5
Compound $\rightarrow$ novel	0	0

out flash called compute. Compute isolation is 98.6% on every seed, against 100% for the lookup. The fusion does not turn a bit-flip into a discovery. On the 80 compound trials it does drop the checksum-first priority. It returns sensor on $72.5 \pm 7.5\%$ of them and compute on $27.5 \pm 7.5\%$, and novel on none. The lookup returns compute on all of them, including the trials whose sensor bit is also true. The accuracy belongs to the fusion, which was allowed to see the four labels. The refusal to preserve a corrupt product belongs to the lookup.

H. A CHECKSUM ON A SMALL CONVOLUTIONAL HEAD

The checksum in (1) reads the linear head. It does not read a filter that sits in front of that head. Table 25 and Fig. 19 put the same identity on a smaller network trained here: four filters of width 5 over the eight background-subtracted detector traces, then a linear readout to GRB, SGR, and solar flare. There is no recurrence. This is not the convolutional-recurrent classifier of John and colleagues [13]. The conv checksum compares the sum of the four filter outputs with

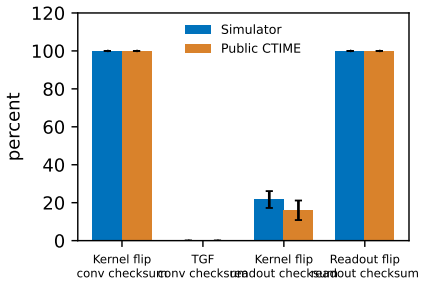


FIGURE 19. Checksum calls from Table 25. The conv checksum catches every critical kernel flip and calls no flash a compute fault. The readout checksum catches its own flips. Its rate on kernel flips is the binary32 leak, not a view of the kernel.

a convolution by the stored sum of those filters. In exact arithmetic the two sides agree for every input, including a flash and a spiked detector. A flipped kernel weight breaks the agreement. The readout keeps its own row-sum checksum, which does not contain the kernels.

Table 25 and Fig. 19 give the rates, including closed-set accuracy $57.2 \pm 1.2\%$ (simulator) and $42.9 \pm 0.5\%$ (CTIME). The readout checksum crosses its clean threshold on $21.7 \pm 4.4\%$ of simulator critical kernel flips and $16.0 \pm 5.1\%$ of CTIME ones. The exact-arithmetic form of that residual stays below 10^{-6}, and no filter output is non-finite. Those trips are the binary32 leak in (1), not a checksum of the kernel.

VII. DISCUSSION

The tables and figures above are the measurement. ProGIP is the nearest range-space neighbor and has no sensor symbol [30]. Structured parity residuals have no symbol for a new class [42]. Checksum-first names a spike plus a bit-flip as compute; a labeled fusion of the same residuals usually names it sensor (Table 24). The four-filter checksum is not the convolutional-recurrent network of John and colleagues [13]. The paper does not report a beam test, a streaming monitor, a martingale test [52], a detector-response template bank, or a classification of catalog particle rows.

TABLE 25. Checksums on a four-filter convolutional head (percent, mean ± standard deviation over five training seeds). This head is not the network of John and colleagues.

Metric	Simulator	Public CTIME
Closed-set accuracy	57.2 ± 1.2	42.9 ± 0.5
Critical kernel flip, conv checksum	100	100
Mild kernel flip, conv checksum	100	100
TGF above the conv checksum	0	0
Clean known above the conv checksum	0.2 ± 0.4	0.3 ± 0.5
Sensor spike above the conv checksum	0	0
Readout flip, conv checksum	0	0
Critical kernel flip, readout checksum	21.7 ± 4.4	16.0 ± 5.1
Readout flip, readout checksum	100	100
TGF above the readout checksum	0	0

VIII. CONCLUSIONS

ARIA is a four-symbol lookup whose residuals are forbidden from seeing the wrong cause. The contribution is the dictionary that keeps discovery as a non-fault symbol.

DATA AVAILABILITY

Fermi-GBM files are public through HEASARC. Scored cubes are `cubes.jsonl`, `cubes_draw2.jsonl`, `tte_cubes_grb_tgf.jsonl`, and `tte_cubes.jsonl`, under `experiments/gbm/`. Channel, checksum, and Poisson counts are `channel_id.json`, `conv_checksum.json`, and `poisson_filter.json`, regenerated from the matching programs in `src/lib/aria/`. Raw FITS files are not included.

AUTHOR CONTRIBUTIONS

Suhasini S. is the corresponding author. **Haripriya S.:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. **Suhasini S.:** Supervision, Writing – review and editing.

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DECLARATION OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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